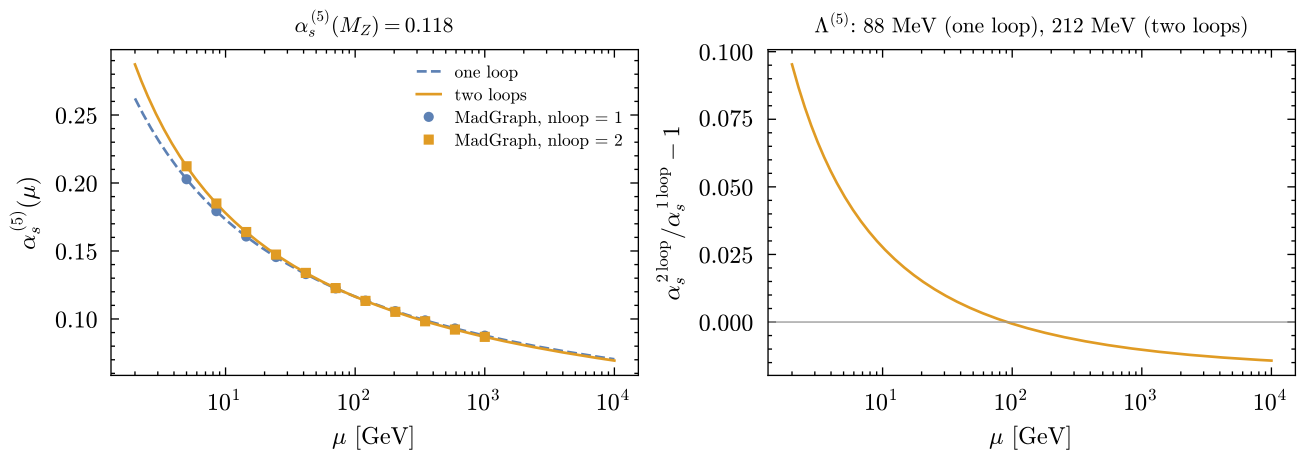
$\alpha_s^{(5)}(M_Z) = 0.118$
one loop
two loops
MadGraph, nloop = 1
MadGraph, nloop = 2
$\alpha_s^{(5)}(\mu)$
0.25
0.20
0.15
0.10
10^1
10^2
10^3
10^4
μ [GeV]
$\Lambda^{(5)}$: 88 MeV (one loop), 212 MeV (two loops)
$\alpha_s^{2\text{loop}}/\alpha_s^{1\text{loop}} - 1$
0.100
0.075
0.050
0.025
0.000
10^1
10^2
10^3
10^4
μ [GeV]